

EEE/CSE 120

Answer Sheet

Capstone Design Project

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Instructor: Goryll

Class Time: 12:00 PM-1:15 PM

Date: 11/21/2019

Task C-1: Design of Synchronous Sequential Machines

Design #1 (Mealy): What assumptions did you make in the design of this machine?

- An assumption I made in the design of this machine was that the player will score three points and then he will win; in addition, I assumed that if the player won the output will be "1" and if he did anything else he will start over.

Create a state definition table here that describes in plain English what each state in your machine means and what binary values you have assigned to represent each state.

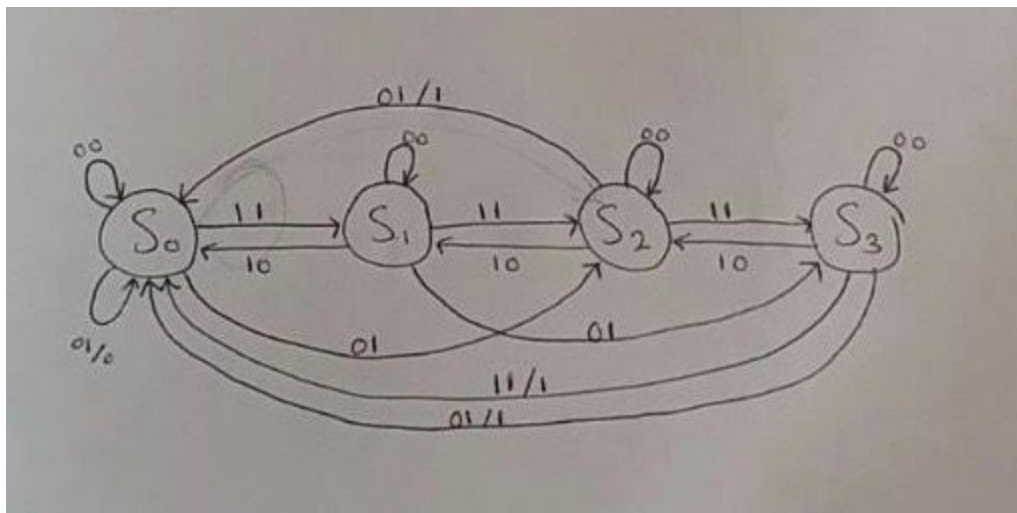
State	English	Binary value
S ₀	Nothing received	00
S ₁	1 score	01
S ₂	Scored 2	10
S ₃	Scored 3	11

if A = 1 hits board if B = 1 hits ring
 A = 0 doesn't hit board B = 0 doesn't hit ring

AB Table

A	B	Score
0	0	0
0	1	+2
1	0	-1
1	1	+1

Create tables here to display your state diagrams, state transition tables and Karnaugh maps used in your design process. (You can do this by hand if you wish.)



Transition table:

A	B	Present state		Next state		Output Y
		Q_0	Q_1	Q_0^+	Q_1^+	
0	0	0	0	0	0	0
0	0	0	1	0	1	0
0	0	1	0	1	0	0
0	0	1	1	1	1	1
0	1	0	0	1	0	0
0	1	0	1	0	0	1
0	1	1	0	0	1	0
0	1	1	1	1	1	1
1	0	0	0	0	0	0
1	0	0	1	0	1	0
1	0	1	0	1	0	0
1	0	1	1	1	1	1
1	1	0	0	1	0	0
1	1	0	1	0	0	1
1	1	1	0	0	1	0
1	1	1	1	1	1	1

K-Maps Q_0^+, Q_1^+, Y

Q_0^+

$Q_0 Q_1$ \ AB	00	01	11	10
00	0	1	0	0
01	0	0	1	0
11	x	x	x	x
10	1	0	0	0

Q_1^+

$Q_0 Q_1$ \ AB	00	01	11	10
00	0	0	1	0
01	1	0	0	0
11	x	x	x	x
10	0	0	0	1

$$Q_0^+ = A'B'Q_0 + ABQ_1 + A'BQ_0'Q_1' \quad Q_1^+ = A'B'Q_1 + AB'Q_0 + ABQ_0'Q_1'$$

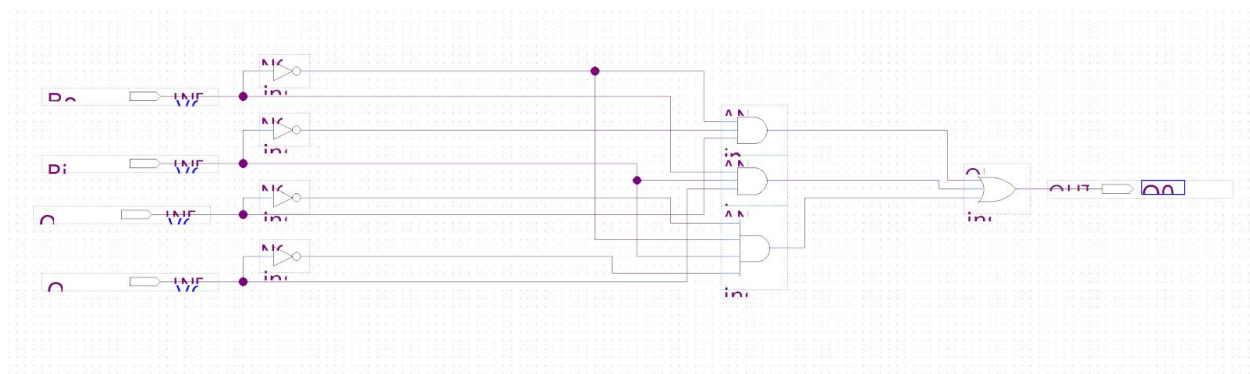
Y

$Q_0 Q_1$ \ AB	00	01	11	10
00	0	0	0	0
01	0	1	0	0
11	x	x	x	x
10	0	1	1	0

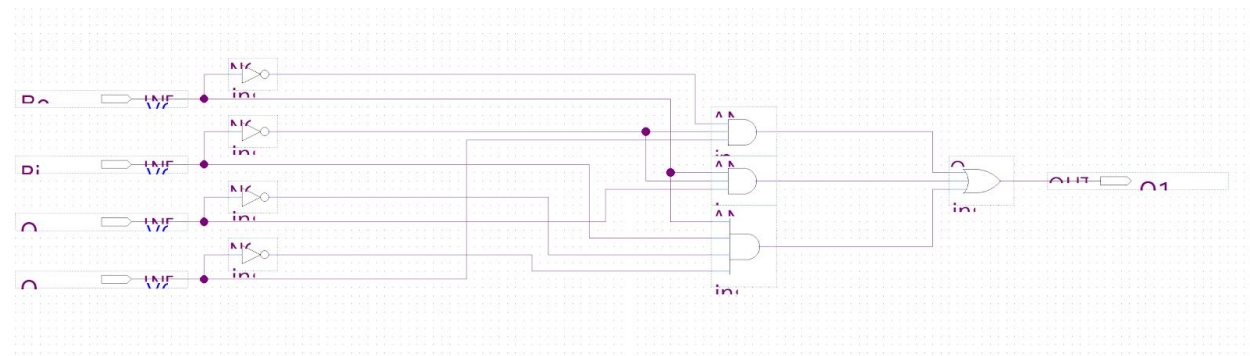
$$Y = Q_0 Q_1' B + A' B Q_1$$

Cut and paste your Quartus schematic and timing diagram simulation for Design #1 here:

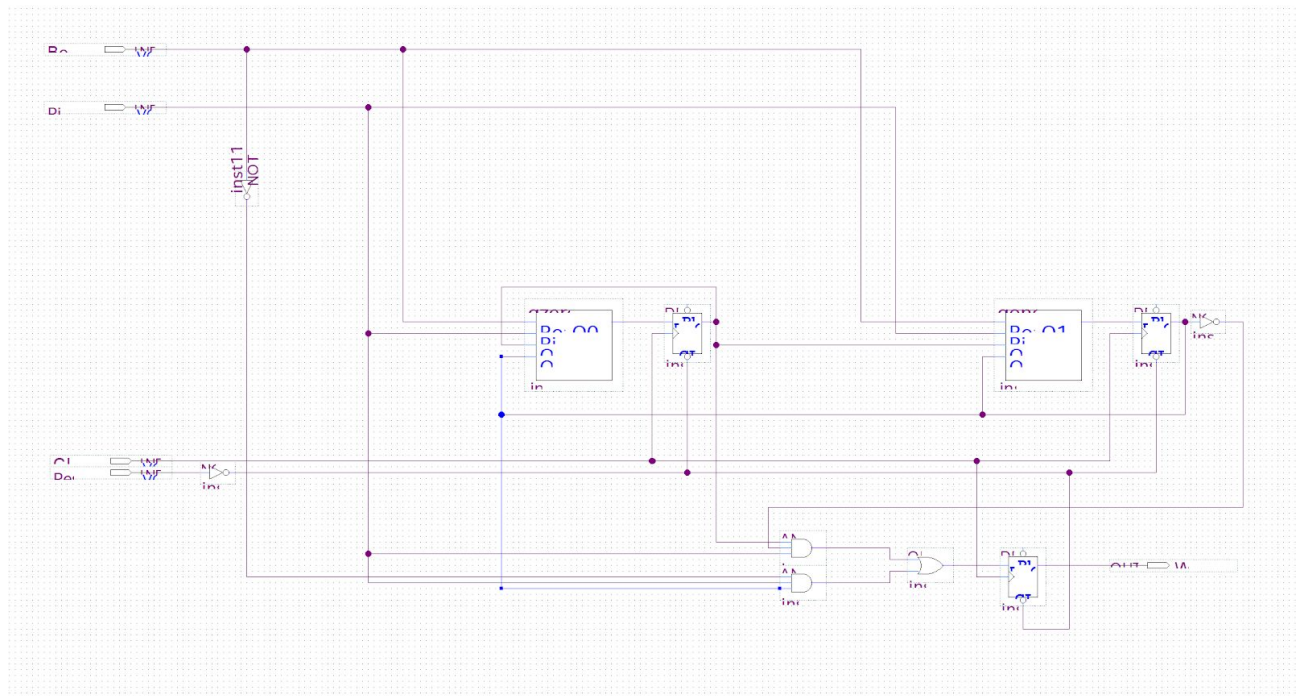
Q0_plus



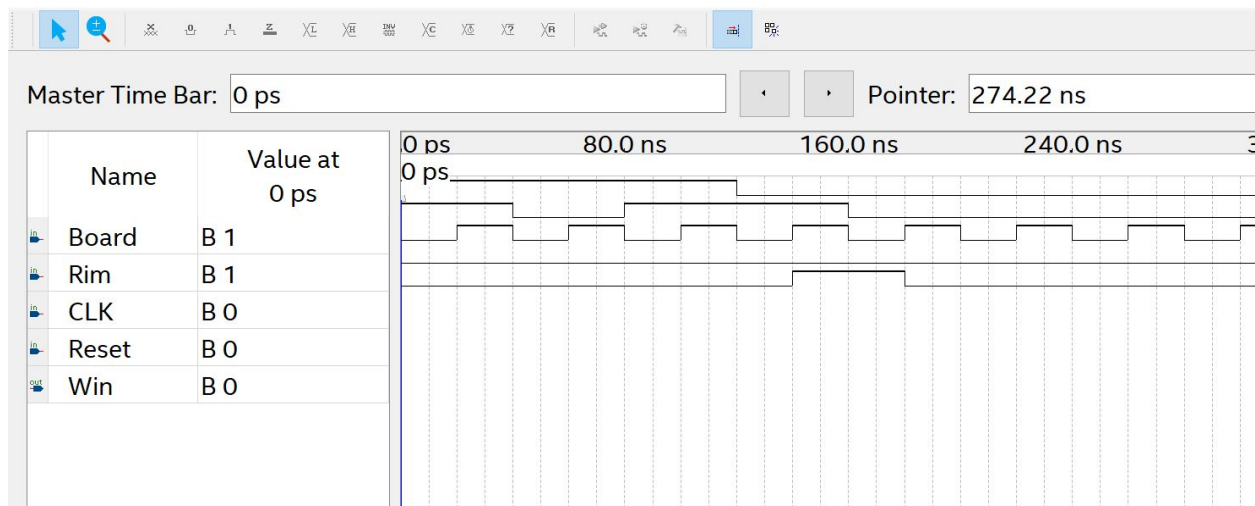
Q1_plus



Mealy machine



Simulation Waveform Editor - C:/Users/Sultan/Desktop/clab/clab - clab - [clab_20191127010134.sim.vwf (Rea
File Edit View Simulation Help



Design #2 (Moore): What assumptions did you make in the design of this machine?

- An assumption I made about this machine was that there were four states, one for each player getting a point, one for each player winning, and one for a tie. Another assumption I made about this machine was that the output was one only if a player won, not if a player gained a point.

Create a state definition table here that describes in plain English what each state in your machine means and what binary values you have assigned to represent each state.

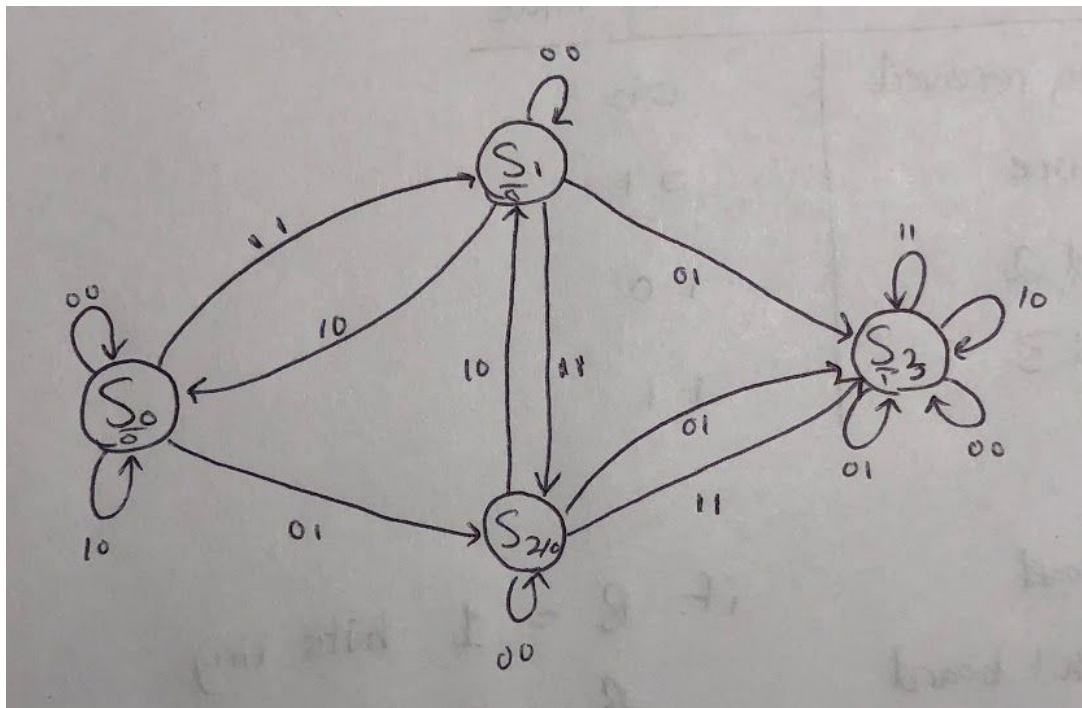
State	Words	Binary value
S ₀	Nothing	00
S ₁	Scored 1	01
S ₂	Scored 2	10
S ₃	Scored 3	11

A	B	Score
0	0	0
0	1	+2
1	0	-1
1	1	+1

A { 1 hits board
0 None

B { 1 hit rim
0 None

Create tables here to display your state diagrams, state transition tables and Karnaugh maps used in your design process. (You can do this by hand if you wish.)



A	B	Present		Next		output Y
		Q_0	Q_1	Q_0^+	Q_1^+	
0	0	0	0	0	0	0
0	0	0	1	0	1	0
0	0	1	0	1	0	0
0	0	1	1	1	1	1
0	1	0	0	1	0	0
0	1	0	1	1	1	0
0	1	1	0	1	1	0
0	1	1	1	0	0	1
1	0	0	0	0	0	0
1	0	0	1	0	0	0
1	0	1	0	0	1	0
1	0	1	1	0	1	1
1	1	0	0	1	0	0
1	1	0	1	1	0	0
1	1	1	0	1	1	0
1	1	1	1	1	1	1

Q_0^+ $Q_0 Q_1$

AB	00	01	11	10
00	0	0	0	1
01	1	1	0	1
11	0	1	0	1
10	0	0	0	0

Q_1^+ $Q_0 Q_1$

AB	00	01	11	10
00	0	1	0	0
01	0	1	0	1
11	1	0	0	1
10	0	0	0	1

$$Q_0^+ = \bar{A} Q_0 \bar{Q}_1 + B Q_0 \bar{Q}_1 + \bar{A} B \bar{Q}_1 + B \bar{Q}_0 Q_1 + \bar{A} B \bar{Q}_0$$

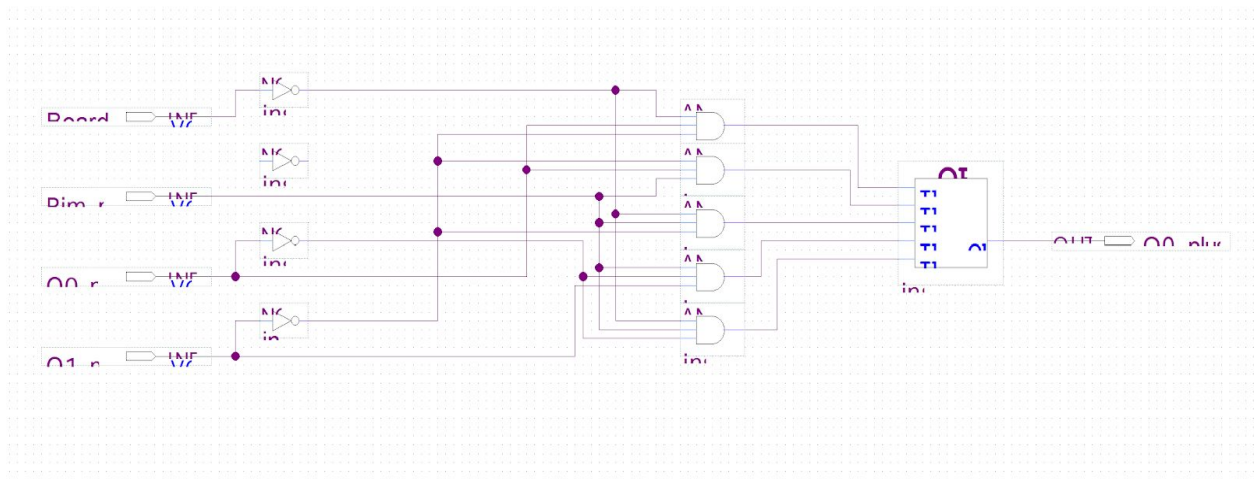
$$Q_1^+ = A \bar{Q}_0 Q_1 + B \bar{Q}_0 Q_1 + A B \bar{Q}_0 + \bar{A} Q_0 \bar{Q}_1$$

Y	$Q_0 Q_1$			
AB	00	01	11	10
00	0	0	1	0
01	0	0	1	0
11	0	0	1	0
10	0	0	1	0

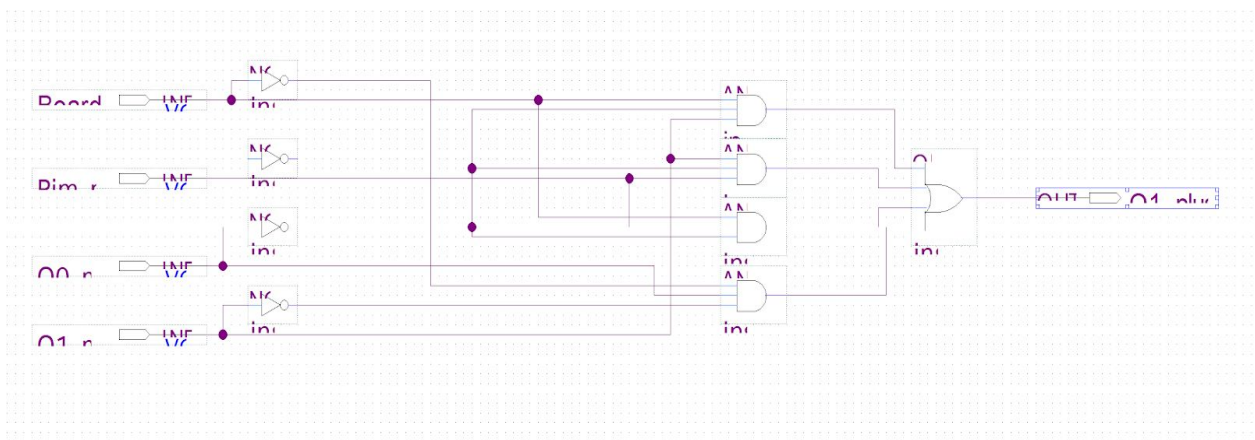
$$Y = Q_0 Q_1$$

Cut and paste your Quartus schematic and timing diagram simulation for Design #2 here:

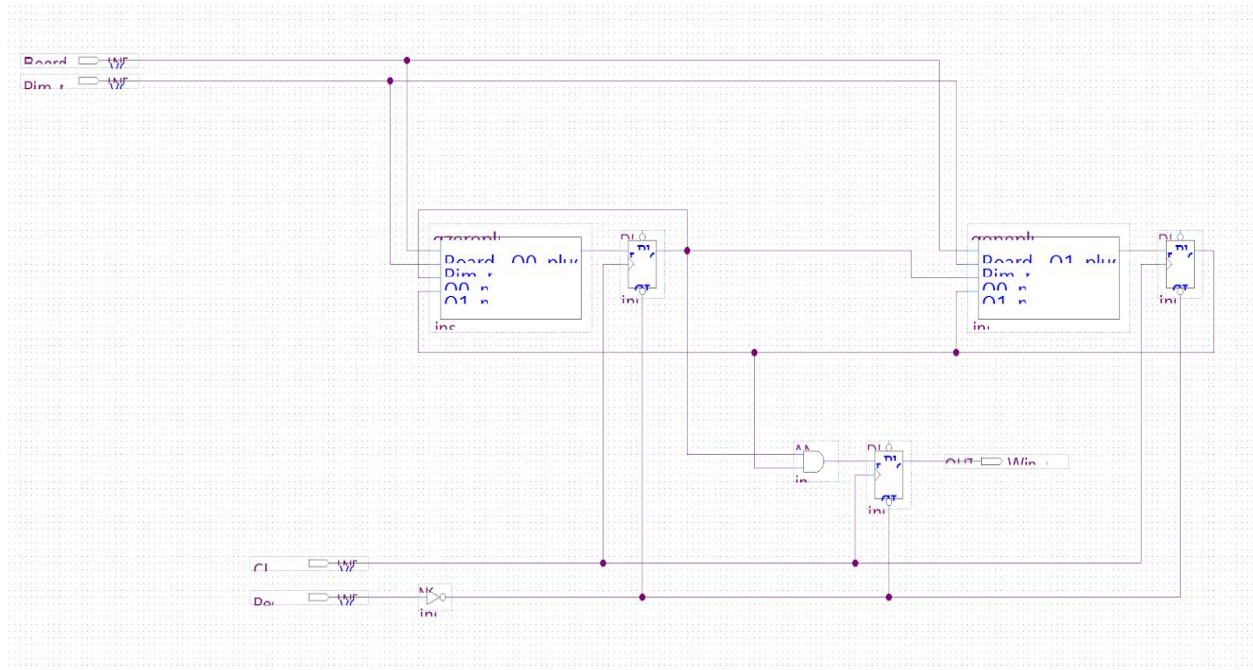
Q0_plus



Q1_plus



Moore machine



Task C-2: Simulate Both Designs in the Lab in Presence of the TA

Demonstrate that both of your circuits meet the completed design specification to the laboratory assistants and have the TAs apply a grade.

Grade: 1st Design ____/15 Grade: 2nd Design ____/15

Task C-3: Determine Criteria and Weighting for Judging Your Designs

Using the guidelines in the laboratory manual, list your criteria and associated weights here:

<u>Criteria</u>	<u>Weight</u>
1. Number of gates	Mealy: 15 Moore: 16
2. Number of flip flops	Mealy: 3 Moore: 3
3. Number of wires	Mealy: 12 Moore: 16
4. Understanding to machine	Mealy: 20 Moore: 20
5. Is it working?	Mealy: 20 Moore: 5
6. Amount of time needed	Mealy: 30 Moore: 40

Task C-4: Apply the Criteria to Pick the Best Design

Describe how you applied the criteria and weighting system in the above task to pick the best design.

- The way I applied the criteria and weighting system was by picking the most important criteria and giving the machines the most points for those criteria and then giving which machine fit the criteria best more points.

Which design is better based on your criteria and weighting system?

- The best design is Mealy machine because it has less wires and it took less time; in addition, it has less number of gates

Task C-5: Upload One Design to the FPGA Board in the Laboratory

Demonstrate to a laboratory assistant that one of your designs works according to your completed specification.

Capstone Design Project: Lab Report Grade Sheet

Name

:

Instructor Assessment:

Grading Criteria	Max Points	Points lost
Template		
Neatness, Clarity, and Concision	5	
Description of Assigned Tasks, Work Performed & Outcomes Met		

Task C-1: Design of Synchronous Sequential Machines	22	
Task C-2: Simulate Both Designs	30	
Task C-3: Determine Criteria and Weighting for Judging Your Designs	5	
Task C-4: Apply the Criteria to Pick the Best Design	8	
Task C-5: Upload One Design to the FPGA Board in the Laboratory	30	
Additional Design Features	(up to 10 extra points)	
Self-Assessment Worksheet (The content of the self-assessment worksheet will not be graded. Full credit is given for including the completed worksheet.)	(2 extra points)	
Lab Score	Points Lost	
	Late Lab	
	Lab Score	

Self-Assessment Worksheet

Put an 'X' in the table below indicating how strongly you agree or disagree that the outcomes of the assigned tasks were achieved. Use '5' to indicate that you 'strongly agree' and '1' to indicate that you 'strongly disagree'. Use 'NA', Not Applicable, when the tasks you performed did not elicit this outcome. Credit will be given for including this worksheet with your lab report. However, your **responses will not be graded**, they are for your instructor's information only.

Table 1: Self-Assessment of Outcomes for the Capstone Design Project Lab.

After completing the assigned tasks and report I am able to:	5	4	3	2	1	NA

Use classical design techniques (i.e., state diagrams, state transition tables, and Karnaugh Maps), to design a synchronous sequential machine starting with a functional specification.						
Make assumptions to complete an incomplete functional specification.						
Write a complete functional specification.						
Build, and debug a synchronous sequential machine.						
Develop reasonable engineering criteria for comparing different designs.						
Apply engineering criteria to select a 'best' design.						

Write below any suggestions you have for improving this laboratory exercise so that the stated learning outcomes are achieved.

- I suggest that at least it shows the beginning steps of the project; however, I figured out how to start with it.